When the Care Team Moves On Before the Chart Does

Detecting silent shifts toward comfort care in the ICU

MIMIC-IV Datathon Analysis

2026-04-11

The question in one paragraph

In the intensive care unit, a patient's "code status" — whether everything will be done to save them, or whether the team will focus on comfort — is supposed to be written in the chart. But clinicians sometimes notice that the team has quietly moved on from a patient, sending fewer labs, writing shorter notes, calling fewer specialists, well before that conversation ever happens on paper. We wanted to know whether that gap between what the team is doing and what the chart says is real, whether we can measure it, and who it happens to.

Why it matters

When the care team's internal picture of a patient differs from what the chart documents, several things can go wrong. Families and patients may still believe the team is fighting for recovery when the team has already shifted toward comfort. Quality reviews and fairness audits, which rely on the chart, miss these patients entirely. And the patient themselves may drift into a softer kind of care without the conversation that would let them choose it.

Our goal was not to find fault with anyone, but to see whether we could make this quiet shift visible in the data.

How we approached it

We used MIMIC-IV, a large public database of real ICU stays from a Boston teaching hospital. About 46,000 adult ICU stays were long enough to study.

The core idea. Some patients in the database formally transitioned from full-code status to a Do-Not-Resuscitate (DNR) or Comfort Measures Only (CMO) order during their stay. We used those patients as a kind of ground truth: we looked at how the *care delivery pattern* changed around the moment of that order, built a numerical "palliative-ness score" that captures the shift, and then asked whether any patients who never got a DNR or CMO order nonetheless show the same pattern.

What goes into the score. Four everyday measures of how intensely a team is engaging with a patient:

- How many lab tests per hour are being ordered
- How *different* those lab tests are (a wide workup versus the same few repeated tubes)
- How much decision-oriented charting is happening (notes beyond routine vitals and alarms)
- How many new specialist consultations are being called

When a team is in rescue mode, all four of these are high. When a team has shifted toward comfort, all four drop.

Adjusting for how sick the patient is. A critical catch here is that a patient who is simply getting better also needs fewer labs and fewer consults. To keep that from looking like a palliative shift, we compare each patient's engagement level to the level we'd *expect* for a patient who is as sick as they are — using their severity scores, whether they are on a ventilator, whether they are on blood-pressure medicines, and similar factors. The score only goes up when engagement is lower than expected for the patient's severity, not when the patient is actually improving.

What we found

1. The shift is real, and it starts before the chart catches up

For about 1,500 patients who did eventually get a DNR or CMO order, we compared three windows within the same patient: a baseline window early in the stay, the day just before the order, and the day after the order. The palliative-ness score rises meaningfully in the pre-order window — before any formal code-status change has happened. About 42% of the total behavioral shift between early-stay rescue care and post-order comfort care has already taken place before the order is written.

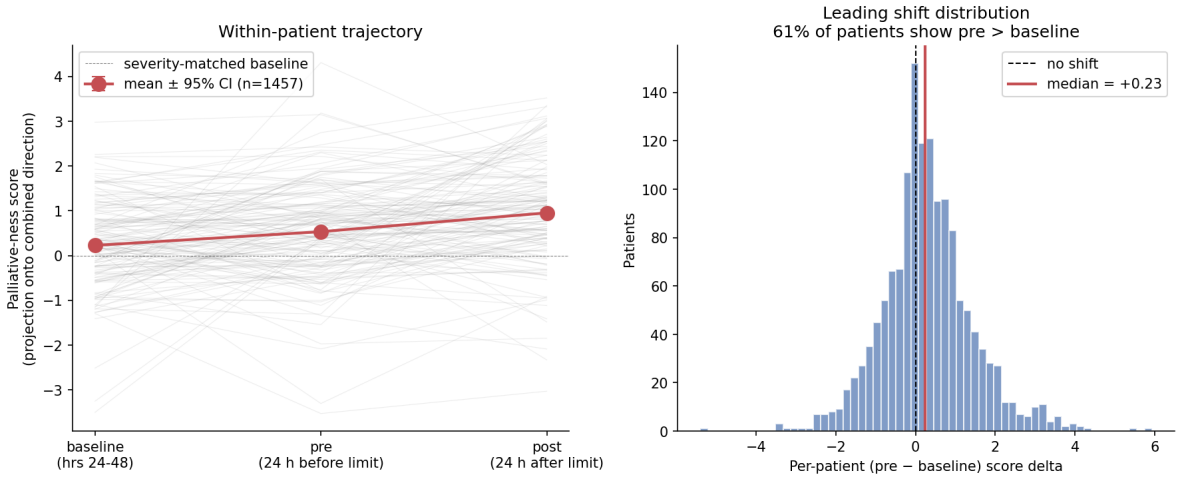

Figure 1: For 1,457 patients who transitioned to DNR or CMO, the palliative-ness score rises before the formal order. Each gray line is one patient, the red line is the average, and the shaded band is the 95% confidence range. The right panel shows how many patients had a positive "leading shift" — about two-thirds of them did.

Because we compared each patient to themselves rather than comparing different patients, this result rules out the obvious alternative explanations: it is not that the patients who eventually got CMO orders had different doctors, or were in different units, or had different baseline illnesses. The same team, treating the same patient, was measurably engaging less in the 24 hours before the order was written.

2. The shift shows up about 48 hours before the formal order

If we line up every transitioning patient so that day zero is the day their DNR or CMO order was written, and then compute the average palliative-ness score on each day before and after, we can see exactly when the signal appears.

Three days out, there is nothing. Two days before the order, the signal first becomes statistically visible. One day before, it jumps sharply. On the day of the order, it peaks. In plain terms: for CMO transitions, the team's behavior starts to soften about 48 hours before the chart is updated.

CMO transitions show a much sharper cliff than DNR-only transitions, which makes sense clinically — CMO is a complete commitment to comfort care, while DNR only limits one specific intervention and is often compatible with ongoing aggressive treatment.

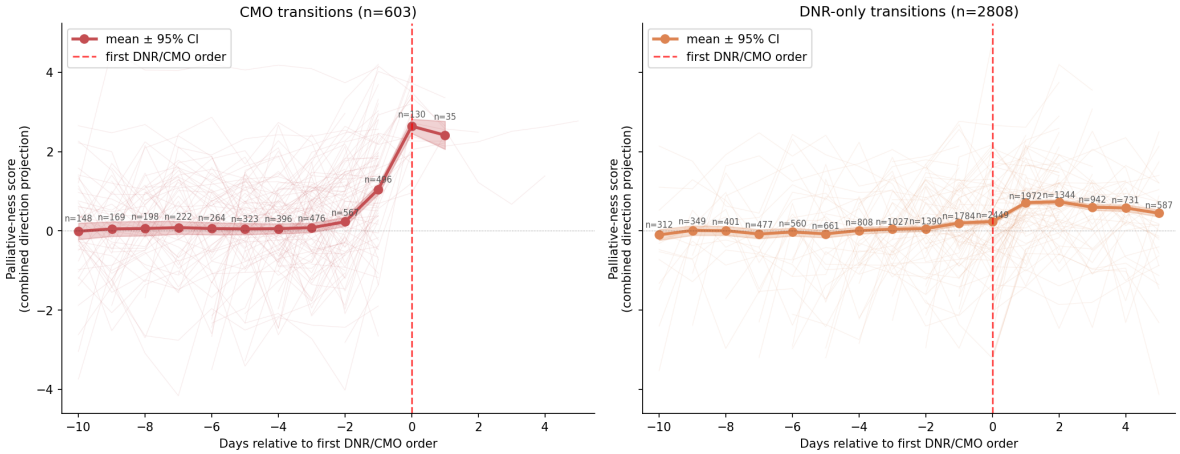

Figure 2: Daily palliative-ness score trajectory for CMO transitions (left, 603 patients) and DNR-only transitions (right, 2,808 patients). The red dashed line marks the day the order was written. CMO patients show a sharp rise about two days before the order; DNR patients show a gentler, more gradual climb.

3. There are patients who never get a formal order but match the pattern

Here is the main question of the study. Among the roughly 43,000 ICU stays that never had a DNR or CMO order, can we find patients whose care pattern matches the palliative-ness signature?

We applied a simple filter. To count as a silent candidate, a patient had to meet all of the following:

1. **Stayed in the ICU long enough to see a trajectory** — at least about three days, so we had multiple 24-hour windows to compare across
2. **Had no formal DNR or CMO order** — this is the primary research population, patients whose chart never changed
3. **Looked like normal rescue-mode care at the start** — their early-stay palliative-ness score was low, meaning we did not include patients who already looked palliative on arrival
4. **Then shifted** — at least two days in a row of clearly elevated palliative-ness score
5. **While still sick** — their severity score during the elevated period was not improving, so the shift could not simply be explained by the patient getting better

Starting from all 94,458 ICU stays in MIMIC-IV, the 48-hour length-of-stay cutoff halves the pool to about 46,500. Dropping patients who ever received a formal DNR or CMO order takes us to about 42,900. Requiring enough ICU time to observe a trajectory leaves about 28,500. Requiring a rescue-mode baseline at admission trims that only slightly to about 26,800 — most long-enough-stay patients do start out looking like rescue care. The sharp cut is the persistence criterion: only about 1,700 patients then show two consecutive days of elevated palliative-ness. Adding the non-improving-severity constraint leaves **1,217 silent candidates**, or about 2.8% of the never-transition population.

For comparison, among patients who did eventually transition to CMO, 6.4% meet the same filter — a 2.2-fold enrichment. This is our validation that the filter is picking up something real rather than random noise.

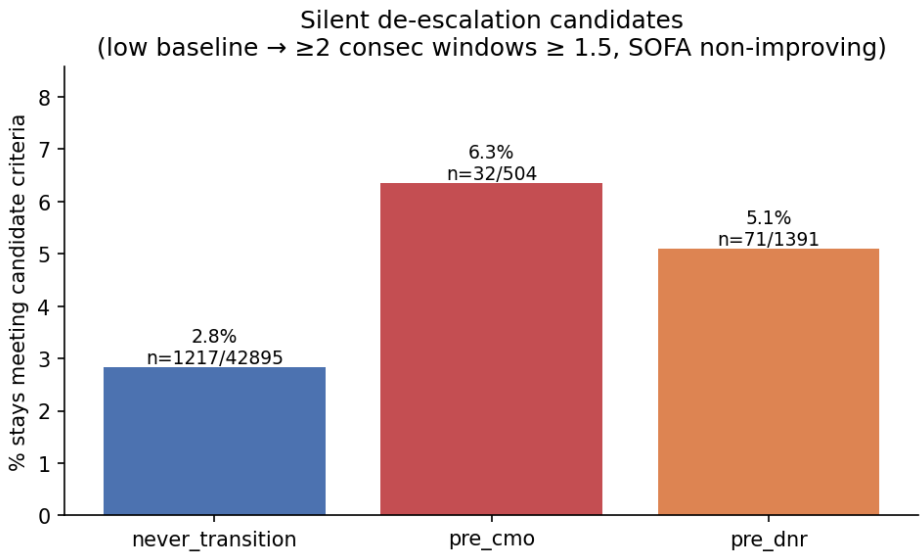

Figure 3: Candidate rate by group. Never-transition stays are the primary research population; the pre-DNR and pre-CMO groups act as positive controls — stays that we already know had a documented shift, used here to validate that the filter really does catch the pattern.

4. The silent candidates are exactly who the clinical team suspected

The clinical intuition going into this study was that silent de-escalation would happen most often in older, sicker, multimorbid patients with long, stop-and-start ICU stays — the patients for whom goals-of-care conversations are hardest to have because there is no single dramatic moment that forces them. The data matches this intuition on every measure we checked.

On the patient-factor side, compared to other full-code patients who were never candidates:

- They are **2.6 times more likely** to have a long ICU stay (three weeks or more)
- They are **1.8 times more likely** to be multimorbid (have many chronic conditions)
- They are **1.4 times more likely** to be 75 or older
- They are **1.3 times more likely** to be repeat ICU users

And on the outcome side:

- They are **2.8 times more likely** to die in the hospital
- They are **1.8 times more likely** to be discharged to hospice
- They are **about 2.5 times less likely** to go home (odds ratio 0.39)

Importantly, the things that did *not* show up as significant are also worth noting. Gender, Black or Hispanic or Asian ethnicity, and marital status were all tested and none of them changed the odds of being a silent candidate after correcting for multiple comparisons. Whatever is driving this pattern, it does not appear to be a straightforward demographic disparity.

The discharge destinations tell the clearest story. Silent candidates are half as likely to go home and twice as likely to die in the hospital as their peers. They are also more likely to be transferred to hospice.

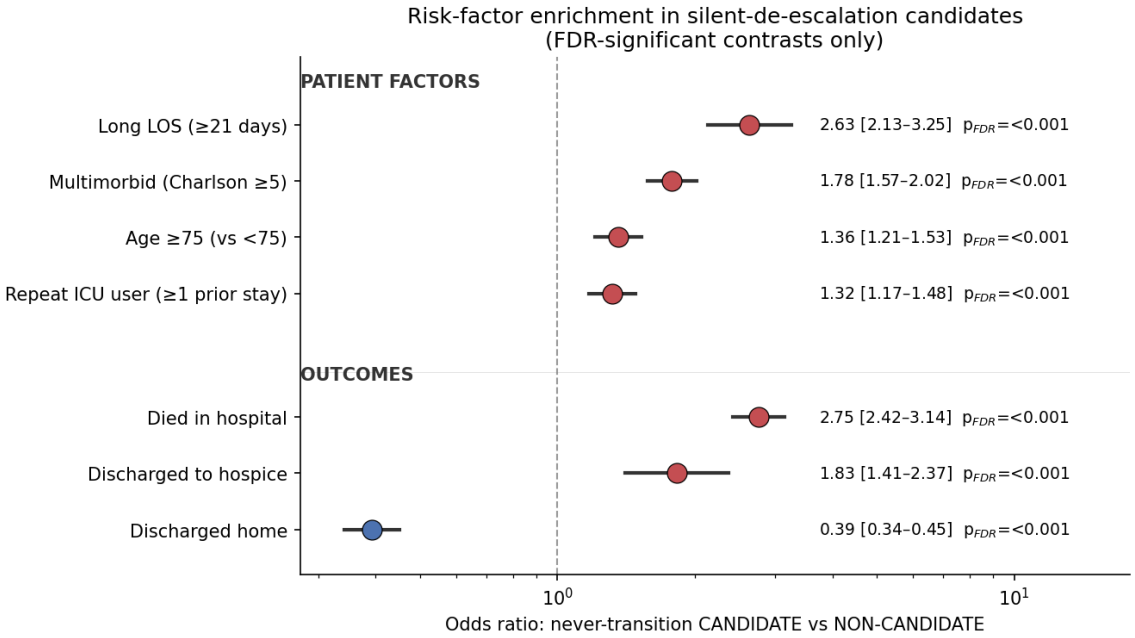

Figure 4: Odds ratios for the statistically significant patient factors and outcomes, candidate versus non-candidate within the never-transition group. The plot is split into two groups: patient factors at the top and outcomes at the bottom. Red markers are contrasts where silent candidates are more common; the blue marker at the very bottom ("discharged home") is a protective contrast — silent candidates are much less likely to go home. Non-significant contrasts like gender, ethnicity, and marital status have been omitted.

Where the patient ended up	Silent candidates	Other full-code patients
Died in hospital	26.9%	11.7%
Discharged to hospice	5.1%	2.9%
Transferred to rehab or long-term facility	46.6%	44.0%
Went home	**21.2%**	**40.7%**

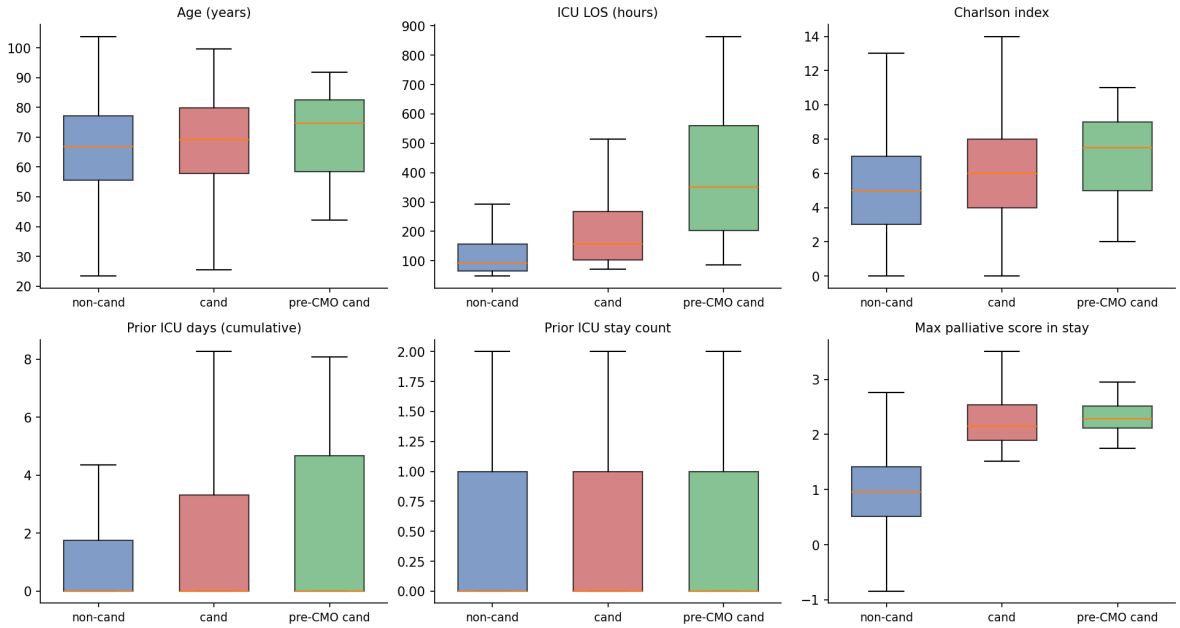

Figure 5: Silent candidates are older, sicker, and stay longer than non-candidates. The third column shows the small group of 32 patients who matched our filter AND eventually had a CMO order — they look directionally similar but more extreme, which gives us additional confidence that the filter is catching a real shift.

What this adds up to

Three independent pieces of evidence all point the same way.

First, when we compare patients to themselves around a DNR or CMO order, we can see the care pattern softening before the order is written. This rules out differences between patients, teams, or hospital units as the explanation.

Second, when we line up the days relative to the order, the softening starts about 48 hours before the formal change and grows over time rather than appearing suddenly.

Third, when we search the larger population of patients who never got an order, we find about 1,200 who match the same signature, and they are exactly the older, sicker, long-stay, multimorbid patients the clinical team suspected. They also have much worse outcomes — fewer of them go home and more of them die in the hospital or are discharged to hospice.

What the study cannot say

Being honest about limits is important.

This is a pattern in data, not a story about any individual patient or clinician. We cannot look at one chart and say "this patient was silently de-escalated." The signal is statistical, and across thousands of patients, it is real — but for any individual patient, the score is one piece of information, not a verdict.

We cannot say the silent shift *caused* the worse outcomes. The patients who match the pattern are sicker to start with, and that might explain most of the outcome difference even if no team had changed its behavior.

We cannot directly observe what the care team was thinking. We are inferring the team's mental picture from how they delivered care. A patient who is simply getting better would also show fewer labs and fewer consults — we tried to control for this by requiring that the severity score is not improving during the silent period, but the control is imperfect.

The analysis was done only in MIMIC-IV, which comes from one academic medical center in Boston. Whether the same pattern appears in other ICUs is unknown.

The ultimate test — whether a clinician reading the notes of one of these 1,217 patients would agree that the care pattern looks palliative — is a chart review, which we have not done.

What this suggests going forward

The value of this analysis is not that it provides a real-time alert or that it identifies specific patients who need different care. It is that it gives researchers, clinicians, and hospital leaders a reproducible way to ask a question that is otherwise very hard to ask: does the care team's behavior sometimes drift away from the chart's stated goals, and if so, for whom?

A few directions this could go:

- **Chart review.** A qualitative review of a random sample of the 1,217 silent candidates would tell us whether clinicians reading the notes would agree that the care pattern looks palliative. This is the critical next step.
- **Fairness analysis.** The demographic breakdowns we captured suggest some differences across age, ethnicity, and marital status. These were not formally tested here, but they could be — and should be, because silent de-escalation is the kind of phenomenon where disparities would be especially worrying.
- **Goals-of-care prompts.** If the signal can be detected prospectively, it could be used to prompt formal goals-of-care conversations earlier, giving patients and families more time to think about what they want and more voice in the decision.
- **External validation.** Checking whether the same pattern appears in other large ICU databases would tell us whether this is a feature of one hospital's culture or a wider phenomenon.

The bottom line

In a large ICU database, we can detect a meaningful behavioral shift toward comfort care that happens roughly two days before the chart is formally updated. The same shift appears in a small but nontrivial group of patients who never have their chart updated at all — about one in every thirty-five full-code patients. These patients are older and sicker, they stay longer, and they are much less likely to go home.

Whether this is a problem to fix or a feature of good care is not ours to say. But it is measurable, it is consistent, and it happens to the patients who are already hardest to care for. That seems worth knowing.

A note on methods

For readers who want the technical details:

Data source. MIMIC-IV version 2.2, accessed via a local DuckDB instance. The cohort is 46,337 adult ICU stays of at least 48 hours. Code status was extracted from chartevents items 223758, 229784, and 228687, with values bucketed as Full Code, DNR, DNI, or CMO.

Palliative-ness score. For every 24-hour window of every stay, four features were computed: labs per hour, distinct lab items per hour, decision-oriented charting events per hour (excluding routine vitals, alarms, respiratory, and lab categories), and new consults per hour. Each feature was residualized against severity covariates (SOFA max and mean, ventilator use, vasopressor use, time into stay, weekend, hour of day, window duration) using ordinary least squares. The residuals were z-scored. A direction vector was computed from the within-patient paired delta (post-transition minus pre-transition residuals) averaged across 1,898 transition stays, then normalized to unit length. The score is the dot product of a window's z-scored residuals with this direction vector.

Silent candidate filter. At least three scored full-code windows, at least one baseline window with score below 0.5 in the first 48 hours, at least two consecutive windows with score of 1.5 or higher after the baseline, and SOFA max at the elevated window at or above SOFA max at the baseline window.

Statistical tests. Wilcoxon signed-rank tests for paired within-patient deltas. Chi-square for categorical comparisons, Mann-Whitney for continuous. Benjamini-Hochberg false discovery rate correction for the characterization table.

A complete technical report with equations, figures for every intermediate step, and cohort flow numbers is available as a companion document.

Reproducibility. All code and intermediate data live at `/Users/ahramhan/Documents/mayo_26/studies/silen`
Scripts run in numbered order from `01_cohort_definition` to `07_characterize_candidates`.